

Critical Analysis of Recent Research Articles

Part 1: Article Choice

Article 1: “Quantifying the dynamic spreading of a molten sand droplet using multiphase mesoscopic simulations” [1].

Justification:

The article was chosen because it emphasizes the hydrodynamic effects of the interaction between molten particles at high viscosities with a substrate. This is important for understanding the spreading dynamics of molten sand particles ingested in a gas turbine engine. This understanding can help improve designs of engine components and thermal barrier coatings. This study is considered under the field of droplet dynamics which combines fluid dynamics aspects such as microfluidics, interfacial flows, and multiphase flows.

Article Summary:

The article aims to study the spreading behavior of molten sand particles or CMAS particles on a solid substrate in isothermal conditions and aims to contrast the spreading behavior of CMAS particles with other highly viscous droplets. The article highlights a few key observations from the interaction of droplets with a solid surface. Namely, the spreading rate is high upon contact because of the inertial and capillary forces present on the droplet. Moreover, the spreading radius of droplets with lower viscosity tends to follow a power-law relationship with time, however the spreading radius for droplets with high viscosity follows a linear relationship with time considering logarithmic corrections. Additionally, the article notes that the spreading rate of a droplet is highly dependent on the viscosity.

The researchers observed that the spreading behavior of the molten sand droplets agrees with the theory and experimental results. They also realized that the simulations concluded that the spreading rate is remained nearly constant in an inertial regime for different equilibrium contact angles. This realization made the researchers conclude that spreading behavior in the early stages from the contact is unaffected by the initial drop size and the equilibrium contact angle.

Article 2: “Spreading dynamics of the viscous droplet impacting on a spherical particle” [2].

Justification:

Like Article 1, Article 2 intends on studying the dynamical behavior of droplets when impacted on a particle. This article was chosen because the authors of this article have measured the spreading effects of low and high viscosity droplets. The authors also studied the collision effects on a curved surface, such as a round particle instead of a solid substrate which has been already extensively studied. This article, like article 1, falls under a multiphase flow problem as well as an instability problem.

Article Summary:

Researchers in this article aim to utilize high-speed imaging techniques to study the spatial and temporal variation of droplets on a target particle surface. They also aim to theoretically comprehend the associated energy and force analyses of the droplet. Lastly, they intend to capture the microscopic and evolutionary behaviors of the droplets in perspective.

The researchers noted that the Weber number is a useful parameter in predicting the maximum spreading area of the droplets. They also realized that the viscosity of the liquid has a direct effect on the liquid-film evolution and the spreading area of the droplets as the trends were different. Lastly, they determined that Weber and Reynolds numbers influenced the droplet's contact angle with the surface, and the droplet to particle size affected the oscillation period of the droplet's impact.

Part 2: Methods Analysis

Article 1: "Quantifying the dynamic spreading of a molten sand droplet using multiphase mesoscopic simulations" [1].

Methods Analysis:

The researchers used a numerical model to measure the spreading radius of CMAS molten particles at different initial drop sizes and equilibrium contact angles. The model the team used agreed well with particles flowing at lower viscosities. However, the model failed to maintain agreement with the real data at higher viscosities due to the limitations put forward by the logarithmic correction they utilized. The team made sure to nondimensionalize many of the used parameters in the model and as known dimensional analysis is a powerful tool to model and scale important parameters. However, the accuracy of dimensional analysis can be limited to fixed geometrical values. Values such as the Reynolds number and the Weber number can have notable discrepancies deeming the solution inaccurate in comparison to realistic results. To improve their model, the team hinted at the implementation of more parameters of interest such as the surface roughness, pressure and so on in their future models. This could increase the complexity of the model but would deem it more reliable. One other method the team can investigate is First Principles methods to obtain more accurate and empirical models.

Article 2: “Spreading dynamics of the viscous droplet impacting on a spherical particle” [2].

Methods Analysis:

The researchers used a theoretical model mainly held by key non-dimensional values including the Weber number, and the Reynolds number amongst others to characterize the collision process of a non-ionized water droplet on a curved surface. The team also used a microscale high speed camera to obtain real time data of the droplet’s interactions with the contacting surface and the spreading behaviors resulting. The team has conducted many error mitigation protocols for the camera system to reduce noise and to minimize the discrepancy in results. However, they have admitted that errors in image post processing affected the crucial geometrical and inertial measurements of the droplets. For example, the droplet, as they said was jittery and not round when it left the syringe, and accounting for that is extremely difficult. The strength of cameras, however, is their ability to very accurately and with minimal complexity track the particle spreading behavior in real time and see clearly where collision effects take place. However, cameras fall short in comparison to numerical and analytical models in terms of accuracy and insight to physical phenomena, and cameras are also more computationally demanding for operation. As discussed before, the theoretical model through dimensional analysis is a powerful tool, but phenomena such as particle impingement on surfaces is difficult to capture accurately by making the usual assumptions that engineers make for other scenarios, thus numerical models, and first principles approaches could be handy in such situations.

Part 3: Full Paper

The articles selected for this report highlight the significance in studying the dynamical effects of droplet interactions within solid and gaseous mediums. This field of research is categorized as droplet dynamics which involves fluid dynamics aspects such as microfluidics, interfacial flow, multiphase flow, and instability problems. One prominent application where this analysis is crucial is in the operation and safety of gas turbines, especially aeroengines. Upon exposure to dusty environments or similar environments where glassy minerals called CMAS are prevalent, aeroengines are prone to catastrophic failure as those particles get ingested into the engine and prolonged past the combustion chamber as molten droplets that deposit on the guide vanes of the turbine stages and thermal barrier coatings (TBCs). This effect significantly degrades the thermal and aerodynamic performance of the engine causing alarming failures from the endured damage. As a result, significant investments from industry and government agencies are made to study the hydrodynamic behavior of those droplets upon contact with a surface at different impact conditions in order to develop more resilient engine components.

The first article investigated, “Quantifying the dynamic spreading of a molten sand droplet using multiphase mesoscopic simulations”, aims to study the spreading dynamics of molten CMAS particles impacting a solid, isothermal substrate. Simulations were performed using the Multiphase Dissipative Particle Dynamics (MDPD) numerical model to capture the spreading rate of a CMAS particle under different contact angles. The simulation results were evaluated against experimental and numerical results obtained for CMAS droplets and other highly viscous droplets and were also contrasted against findings for fluids of lower viscosity.

Some of the key observations that the article highlighted from the interaction between droplets and a solid surface were that the inertial and capillary forces experienced on the droplet upon impact cause the droplet to maintain a higher initial spreading rate. Moreover, the spreading radius of droplets with lower viscosities follows a power-law relationship with time, however, the spreading radius follows a linear relationship with time for droplets with higher viscosities after applying logarithmic corrections.

The researchers realized that the obtained numerical results for the spreading behavior of molten CMAS droplets agree well with theoretical and experimental observations. Confirming previous research, the researchers observed that the spreading rate of low-viscosity droplets is directly affected by the equilibrium contact angle. Smaller equilibrium contact angles yielded a larger spreading rate while larger equilibrium contact angles yielded a smaller spreading rate. However, they noted that the spreading rate of droplets with high viscosities such as CMAS remains constant for different equilibrium contact angles because of viscous dissipation effects. Such observation led the researchers to conclude that the spreading behavior of CMAS droplets in the early (inertial) stages of the impingement process is unaffected by the droplet’s initial drop size and the respective equilibrium contact angle.

The second article selected, “Spreading dynamics of the viscous droplet impacting on a spherical particle”, aims to study the collision effects of droplets on smooth particles using theoretical analysis to model the forces and energy conversions occurring from the impingement process and an experimental approach through the utilization of high-speed photography to capture the spreading characteristics of the droplet at different impact conditions. Crucial non dimensional parameters such as the Weber number, the Reynolds number (Re), and the droplet-to-particle (DTP) size ratio were varied to study the evolution of the dimensionless spreading radius, the dimensionless liquid film thickness, and the dynamic contact angle.

Some of the key observations that the researchers noted included the realization that the spreading area of droplets with lower viscosity such as de-ionized water increases to a maximum upon impingement and gradually decreases after some time. However, the spreading area of a highly viscous droplet made from glycerin, for example, remains constant after increasing to a quasi-equilibrium state. They also noted that the viscosity of the droplet has a notable effect on the liquid film evolution and the associated spreading area as seen by the differing trends in the results. Moreover, they realized that the Reynolds number and the Weber number had the greater influence in determining the droplet’s dynamic contact angle upon the surface, while the droplet-to-particle size ratio had the greater effect in characterizing the oscillation period of the collision.

The researchers in the first article utilized a numerical model to analyze the evolution of the spreading radius of molten CMAS particles impinging on a smooth aluminum substrate at different initial drop sizes and contact angles at isothermal conditions. Critical parameters such as the density, surface tension, and dynamic viscosity of the droplet were mapped to equivalent parameter scalars understood by the model. The Multiphase Dissipative Particle Dynamics (MDPD) numerical model was chosen for this analysis due to the minimized complexity and retained accuracy in capturing the hydrodynamic behavior and rheological nature of soft particles. The MDPD model was specifically adopted over other particle and continuum-based models because the constitutive relations are already driven from the model itself and the user is not required to specify these relations. This allows the MDPD model to capture the complex behavior of non-Newtonian fluids such as molten CMAS droplets more accurately and efficiently. While the model captures the spreading radius well for droplets at lower viscosities in relation to experimental results, it fails to accurately capture the spreading radius at higher viscosities when the logarithmic corrections are applied due to the lack of ample resolution required to capture the evolution of the contact line in the earlier stages of the impingement. Parameter mapping and dimensional analysis are powerful tools implemented to model and scale relevant parameters; however, the scaling accuracy is more directed to fixed physical parameters such as the diameter and density. Scalability of non-dimensional values such as the Reynolds number and the Weber number leads to notable discrepancies that must be corrected to obtain more accurate results. To improve the accuracy of the results, the authors hinted towards a fundamental understanding and implementation of other parameters of interest including the heat transfer coefficient, surface roughness, and the pressure in their future studies. While this approach may lead to an increase in modeling complexity, the results obtained would give better insight of the physical nature of droplet dynamics and would be deemed more reliable. Other possible approaches include the integration of First Principles comprehension with some aspects of the current model to create a more reliable model, or the development of an experimental method using high resolution imaging or laser tracking to accurately capture the spreading behavior in real time.

The researchers in the second article developed a theoretical model held by key non-dimensional parameters including the Weber number, the Reynolds number, and the droplet-to-particle size ratio amongst others to characterize the collision effects, manifested in the associated force and energy distributions, between droplets of differing viscosity and smooth particles made from stainless steel. The team also utilized a microscale, high-speed camera to capture the collision effects and the spreading behavior of the impinging droplet in real time. Those effects were interpreted in terms of the spreading radius, the liquid film thickness, and the dynamic contact angle as discussed before. The researchers have conducted many error mitigation protocols for the camera setup to reduce generated noise, and vibration, minimizing the discrepancy in results obtained in the image post-processing phase. They also realized that other factors related to the droplet itself must be accounted for to obtain accurate results. For example, when the droplet has a low surface tension or when the droplet is just released from the syringe, it exhibits jitter characteristics and is difficult to maintain in its spherical form leading to increased complexity in finding the radius and center of gravity which are needed in evaluating the droplet's velocity. Mitigating these errors can be done through increasing the image's resolution, and by further image preprocessing, which is time consuming and demands more computational power. Despite these facts, cameras are still widely used and relied on for analyzing the dynamic behavior of droplets. The strength of cameras lies in their ability to very accurately and with minimal complexity track the particle spreading behavior in real time and see clearly where collision effects take place. However, numerical models, while being more complex to construct, allow for a more nuanced understanding of

the underlying physics in droplet interactions and offer more improved accuracies in meeting expected results. To increase the certainty in results, one approach would be integrating a reliable numerical model in the study and comparing the experimental results with the integrated model.

The researchers in the first article outlined their findings for the evolution of the spreading behavior of molten CMAS droplets in the light of key parameters including the spreading radius and spreading rate of the droplet and the associated contact angle of the droplet. Those parameters were investigated in the inertial and viscous phases of the spreading evolution and were evaluated against different equilibrium contact angles and initial drop sizes. To better characterize the evolution of the spreading behavior for CMAS droplets, the researchers have scaled the spreading radius in the viscous timescale using well-established scaling laws for inertia-dominated and viscous-dominated regimes. The researchers have attributed this decision to the large Ohnesorge (Oh) number of CMAS particles. The Oh number is a dimensionless parameter dictating the ratio between the viscous forces to the inertial and surface tension forces. A plot of the spreading radius evolution over the viscous timescale for CMAS droplets shows a collapse in the data points for different initial drop sizes at a fixed equilibrium contact angle. The collapse was maintained in both the inertial and the viscous regimes. The researchers concluded that the spreading radius is independent of the initial drop size for the entire spreading process. They further validated this result by extracting the spreading rates of the inertial and viscous phases of the graph by curve fitting techniques and then comparing the resulting values with other works and established mathematical laws. This observation is important because it helps better understand how the initial drop size influences the process of droplet solidification through the spreading aspect of the droplet. The researchers observed, however, that changing the equilibrium contact angle influences the spreading radius of the droplet only in the viscous regime since the curve fits well in the inertial regime despite the absence of a collapse in the data. The researchers indicated that the divergence in the spreading radius introduced is due to the effects of partial wetting. The researchers have not provided any insights on the physical behavior involved in partial or complete wetting and so an understanding of this process could be a promising direction in future research endeavors to better understand the spreading behavior of CMAS particles. Moreover, when the researchers plotted the spreading rate against the nondimensional spreading radius for different initial drop sizes, they observed that the curve fitting model they utilized using logarithmic correction matched well with experimental data at lower viscosities while showing a notable diversion at higher viscosities. This deviation resulted from the lack of sufficient resolution to capture the very early time evolution of the spreading process. Lastly, the researchers realized an agreement in the simulation results and established theoretical models for the evolution of the dynamic contact angle at different equilibrium contact angles throughout the spreading process. As discussed previously, in the future, the researchers aim to better understand the interfacial dynamics involved in CMAS interactions by studying the effects of the surface roughness, pressure, phase change, and heat transfer on the spreading radius evolution.

The researchers in the second article outlined their findings for the collision effects of viscous and non-viscous droplets with solid particles. They tracked and modeled the evolutions of the dimensionless spreading radius, the dimensionless liquid film thickness, and the dynamic contact angles through interchanging key parameters such as the Weber number, the Reynolds number, and the droplet-to-particle size ratio. The researchers outlined their results through different plots where the aforementioned parameters were introduced to tackle multiple aspects of the spreading process of the droplet in the evolution. Similar to the process upheld by the researchers in the first article, the

researchers in the second article validated their results through comparing results with established theoretical laws and models, and with other relatable experiments. Additionally, the researchers used multiple curve-fitting techniques based on theory to better model their results and understand the relationships established between the plotted parameters more succinctly. Some of the key findings observed by the researchers include the fact that the dimensionless spreading area of a low-viscosity droplet initially increases to a maximum value then slowly decreases with time, while the dimensionless spreading area of a high-viscosity droplet remains stable after increasing to its maximum value. This observation is crucial for many applications such as in droplet dynamics within gas turbine engines because knowing the spreading area achieved by a molten CMAS particle would help predict the probability of deposition on the occupied surface. Moreover, low-viscosity droplets with low Weber numbers endure significant oscillation periods and the liquid film's oscillation is limited with high Weber numbers and low Reynold numbers. This observation is useful for understanding the effect of the droplet's impact velocity, for example on the oscillatory behavior of the impinging droplet, which can influence the corresponding spreading behavior. It is also noted that the liquid film thickness is minimized when the droplet-to-particle size ratio, Weber number, and Reynold's number are maximized. The dynamic contact angle also changes periodically for low-viscosity droplets when the Weber is small, however, for high-viscosity droplets, the dynamic contact angle decreases until the droplet reaches its equilibrium state in the spreading process. Lastly, the researchers also highlighted the fact that the dynamic contact angle of highly viscous droplets is larger than low-viscosity droplets since the maximum spreading area is limited in the case of high-viscosity fluids. The researchers indicate that the current investigations of the dynamic contact angle are not adequate in properly measuring its value amidst the difference stresses and forces experienced in the spreading process, and they therefore aim to better characterize the dynamic angle and its influencing factors in the future.

The two articles selected for this study are relevant in the field of droplet dynamics. Multiple works have been committed to new numerical and analytical methods to better understand the behavior of droplets upon impacting different surfaces at different conditions. The first article mainly highlighted the promising application of the MDPD simulation model which is capable of accurately modeling the behavior of non-Newtonian fluids. Most models in previous works were limited to Newtonian fluids since those models required the user to construct the accompanying constitutive relations is a time-consuming and complex task. The MDPD model, as discussed previously, integrates the underlying theory, and allows for the analysis of very complex scenarios, which makes it promising to utilize in future studies. The second article allows for a broader understanding of the impact behavior of droplets on curved surfaces such as particles. This allows for an improved understanding of the spreading dynamics of the particle and allows for better models to be constructed. Most studies focused on the impingement of particles upon a flat surface, and few studies deviated from such format.

The field of droplet dynamics is involved in advancing many industries and scientific disciplines. Some of the notable disciplines affected by the study of droplet dynamics include the aerospace industry where molten sand particles interact with the engine. Additionally, the fields of agriculture and environmental science are affected by aspects such the spraying of fertilizer, the formation of rain and soil erosion amongst many others. Another promising study of droplet dynamics is involved in the general field of microfluidics which is crucial in many biological and chemical applications.

In conclusion, the investigated articles discussed in this report provided an improved understanding of the behavior of droplets and their interactions across different phases and mediums. The first article provided valuable insight on the potential of a newly explored numerical model called MPDP which offers the ability to model and study the behavior of non-Newtonian fluids, which is the adopted behavior of molten CMAS droplets. The acquired understanding is crucial for the aviation industry where significant investments are made to improve the safety and efficiency of jet engines. The second article opened a new door for exploring the effects of droplet impingement upon solid particles. Through modeling the spreading behavior of the various droplets of differing material and geometrical characteristics, the researchers were able to give valuable knowledge that can be built upon in future research endeavors to advance many of the industries and disciplines mentioned including materials engineering , pharmaceutical sciences and many more.

References (IEEE)

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